

Dental Clinic Management System

System Analysis and Design Document

Course: System Analysis and Design

Project Type: System Analysis + UI Design + Prototype

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1. Project Introduction

1.1 Overview

In many dental clinics, key processes such as appointment scheduling, patient file management, treatment history tracking, payment recording and daily reporting are still performed manually or in a semi-traditional manner. This leads to errors, wasted time, scheduling conflicts and difficulty in tracking patient histories.

This document presents the full system analysis and design for a Dental Clinic Management System — a web-based platform designed to digitize and streamline all core clinic operations. The system is designed as an expandable initial version that can be developed into a fully functional application in future software engineering courses.

1.2 Project Objectives

- Digitize the main processes of a dental clinic
- Reduce errors in appointment scheduling and patient data entry
- Improve the experience of the secretary, dentist and clinic manager
- Provide a solid foundation for full code implementation in future semesters
- Deliver a complete, presentable prototype of the user interface

1.3 Reason for Topic Selection

This topic is ideal for a system analysis course because it has real and clearly defined users, analyzable and modeled processes, clear functional and non-functional requirements, and the potential for full development in future software engineering courses.

1.4 Project Scope — Initial Version

In the first phase, the system focuses on the following core areas:

- Secretary Dashboard — view and manage today's appointments
- Patient Registration and Search
- Patient Status Management — Waiting, In Room, Visit Completed
- Payment Recording and Daily Income Summary
- Doctor Panel — treatment history and prescription management
- Patient Portal — online appointment booking
- Manager Reports — financial and operational summaries
- Notification System — SMS and email reminders

Features such as database connectivity, real authentication backend, file uploads and SMS gateway integration are noted as future development items and are designed with extensibility in mind.

2. System Stakeholders and Users

2.1 Secretary

The secretary is the primary internal user of the system. They are responsible for registering new patients, scheduling and managing appointments, updating patient visit status throughout the day, recording payments, and viewing the daily dashboard summary. The secretary interacts with the system most frequently and requires a fast, simple interface with minimal clicks.

2.2 Dentist

The dentist accesses the system to view their daily schedule, review patient treatment history before each visit, record services and treatments performed, issue prescriptions, and view important medical alerts such as drug allergies or special conditions. The dentist needs clear, organized access to patient medical data.

2.3 Clinic Manager / Owner

The manager oversees the clinic's financial and operational performance. They use the system to view daily income reports, monitor the number of patients and services, review payment breakdowns, manage staff accounts, and generate monthly financial reports. The manager requires comprehensive reporting and administrative controls.

2.4 Patient

While patients are not direct internal users of the core system, they interact with the system through the patient portal. Patients can register online, book and cancel appointments, view their treatment history and payment records, and receive notifications. All system services are ultimately designed to improve the patient experience.

3. Problems with Traditional Methods

The following problems were identified in clinics that rely on manual or semi-traditional methods:

- Paper patient files may be lost, incomplete or illegible
- Difficulty in quickly searching patient histories
- High risk of scheduling conflicts and double bookings
- Difficulty tracking the current status of each patient during the day
- No quick access to daily income and patient count reports
- Heavy reliance on the secretary's memory and handwritten notes
- Inability to send automated reminders to patients
- No online booking option for patients
- No centralized view of prescriptions or treatment plans
- Manual billing calculations prone to error

4. Functional Requirements

Functional requirements define what the system must do. Each requirement is written using the standard format: “The system shall...” The following table lists all functional requirements organized by module.

ID	Module	Requirement	Priority	Category
FR-01	Secretary	The system shall display a dashboard showing today's full appointment list upon login.	High	Must Have
FR-02	Secretary	The system shall allow the secretary to search for a patient by full name.	High	Must Have
FR-03	Secretary	The system shall allow the secretary to search for a patient by national ID number.	High	Must Have
FR-04	Secretary	The system shall allow the secretary to register a new patient by entering name, national ID, phone number, date of birth and medical conditions.	High	Must Have
FR-05	Secretary	The system shall allow the secretary to change a patient's visit status between Waiting, In Doctor's Room, and Visit Completed.	High	Must Have
FR-06	Secretary	The system shall update the patient status badge immediately on the screen without requiring a page refresh.	High	Must Have
FR-07	Secretary	The system shall allow the secretary to record the payment amount made by a patient after their visit.	High	Must Have
FR-08	Secretary	The system shall display daily summary statistics including total appointments, patients waiting, completed visits, and daily income.	High	Must Have
FR-09	Secretary	The system shall display a log of recent activities on the dashboard such as new registrations and status changes.	Medium	Should Have
FR-10	Secretary	The system shall display a daily checklist of important tasks for the secretary.	Medium	Should Have
FR-11	Secretary	The system shall allow the secretary to contact a patient directly from the appointment list.	Low	Could Have
FR-12	Secretary	The system shall display a patient's important medical alerts such as allergies before their appointment.	High	Must Have
FR-13	Secretary	The system shall allow the secretary to add notes to a patient's file during their visit.	Medium	Should Have
FR-14	Secretary	The system shall allow the secretary to open a patient's full file to view all their information.	High	Must Have
FR-15	Secretary	The system shall track pending payments and display which patients have outstanding balances.	High	Must Have
FR-16	Secretary	The system shall generate a payment receipt for a patient after recording their payment.	Medium	Should Have
FR-17	Authentication	The system shall allow all users to log in using their registered email and password.	High	Must Have
FR-18	Authentication	The system shall redirect users to their role-specific dashboard after successful login.	High	Must Have
FR-19	Authentication	The system shall allow users to log out and terminate their session.	High	Must Have
FR-20	Authentication	The system shall allow users to reset their password by requesting a reset link to their registered email.	High	Must Have
FR-21	Authentication	The system shall lock an account after 5 consecutive failed login attempts.	High	Must Have

ID	Module	Requirement	Priority	Category
FR-22	Authentication	The system shall automatically log out a user after 30 minutes of inactivity.	High	Must Have
FR-23	Authentication	The system shall allow users to change their password from their profile settings.	Medium	Should Have
FR-24	Authentication	The system shall allow users to view and edit their profile information.	Medium	Should Have
FR-25	Authentication	The system shall allow the manager to create new staff accounts and assign roles.	High	Must Have
FR-26	Authentication	The system shall allow the manager to deactivate a staff member's account.	High	Must Have
FR-27	Patient Records	The system shall allow the secretary to register a patient's special medical conditions including diabetes, heart disease and drug allergies.	High	Must Have
FR-28	Patient Records	The system shall store and display the full patient file including contact details, medical history and previous visits.	High	Must Have
FR-29	Patient Records	The system shall display a prominent alert when a patient has drug allergies or special medical conditions.	High	Must Have
FR-30	Patient Records	The system shall maintain a complete history of all treatments performed for each patient.	High	Must Have
FR-31	Patient Records	The system shall allow authorized staff to add notes to a patient's file at any time.	Medium	Should Have
FR-32	Patient Records	The system shall store a minimum of 5 years of treatment history per patient.	High	Must Have
FR-33	Doctor Panel	The system shall allow the dentist to view a patient's full treatment history before starting a visit.	High	Must Have
FR-34	Doctor Panel	The system shall allow the dentist to record the type of treatment or service performed during each visit.	High	Must Have
FR-35	Doctor Panel	The system shall display clear warnings about a patient's drug allergies and sensitivities to the dentist.	High	Must Have
FR-36	Doctor Panel	The system shall allow the dentist to view their schedule for the current day.	High	Must Have
FR-37	Doctor Panel	The system shall allow the dentist to create and save a treatment plan for a patient.	High	Must Have
FR-38	Doctor Panel	The system shall allow the dentist to issue a prescription for a patient after their visit.	High	Must Have
FR-39	Doctor Panel	The system shall display all prescriptions previously issued to a patient.	High	Must Have
FR-40	Doctor Panel	The system shall allow the dentist to print or download a patient's prescription.	Medium	Should Have
FR-41	Doctor Panel	The system shall allow the dentist to request a radiography or OPG image for a patient in future versions.	Low	Could Have
FR-42	Patient Portal	The system shall allow a new patient to register an account online using their name, email and password.	High	Must Have
FR-43	Patient Portal	The system shall allow a registered patient to log in to the patient portal.	High	Must Have
FR-44	Patient Portal	The system shall allow a patient to book an appointment by selecting a doctor, date and available time slot.	High	Must Have

ID	Module	Requirement	Priority	Category
FR-45	Patient Portal	The system shall display only available time slots when a patient is booking an appointment.	High	Must Have
FR-46	Patient Portal	The system shall allow a patient to view their list of upcoming appointments.	High	Must Have
FR-47	Patient Portal	The system shall allow a patient to cancel an upcoming appointment online.	High	Must Have
FR-48	Patient Portal	The system shall allow a patient to view their past treatment history.	Medium	Should Have
FR-49	Patient Portal	The system shall allow a patient to view their payment history and outstanding balance.	Medium	Should Have
FR-50	Patient Portal	The system shall allow a patient to update their personal and medical information.	Medium	Should Have
FR-51	Patient Portal	The system shall send a confirmation message to the patient after a successful booking.	High	Must Have
FR-52	Patient Portal	The system shall allow a patient to view prescriptions issued by their dentist.	Medium	Should Have
FR-53	Patient Portal	The system shall allow a patient to rate their visit experience after an appointment.	Low	Could Have
FR-54	Notifications	The system shall send an SMS reminder to the patient 24 hours before their appointment.	Medium	Should Have
FR-55	Notifications	The system shall send an email confirmation to the patient immediately after booking.	Medium	Should Have
FR-56	Notifications	The system shall send a payment reminder to patients with an outstanding balance.	Medium	Should Have
FR-57	Notifications	The system shall notify the secretary when a new appointment is booked online.	High	Must Have
FR-58	Notifications	The system shall notify the patient when their appointment is confirmed by the clinic.	High	Must Have
FR-59	Notifications	The system shall notify the patient if their appointment is cancelled by the clinic.	High	Must Have
FR-60	Notifications	The system shall send a follow-up reminder to patients after their treatment.	Low	Could Have
FR-61	Notifications	The system shall allow the manager to enable or disable specific notification types.	Medium	Should Have
FR-62	Manager	The system shall allow the manager to view the daily income report.	High	Must Have
FR-63	Manager	The system shall allow the manager to view the daily number of patients and services performed.	High	Must Have
FR-64	Manager	The system shall display a breakdown of payments by type including cash and card.	Medium	Should Have
FR-65	Manager	The system shall display a list of patients with outstanding balances.	Medium	Should Have
FR-66	Manager	The system shall allow the manager to generate a monthly financial report.	Medium	Should Have
FR-67	Manager	The system shall allow the manager to view all insurance claims submitted in a given month.	Medium	Should Have
FR-68	Appointment	The system shall prevent two patients from booking the same time slot with the same doctor.	High	Must Have

ID	Module	Requirement	Priority	Category
FR-69	Appointment	The system shall display the full patient journey from arrival to payment completion.	High	Must Have
FR-70	Appointment	The system shall allow the secretary to add insurance information to a patient's file.	High	Must Have
FR-71	Appointment	The system shall automatically calculate the patient's payment after applying their insurance coverage.	High	Must Have
FR-72	Appointment	The system shall display a confirmation dialog before performing critical actions such as cancellation or deletion.	High	Must Have

5. Non-Functional Requirements

Non-functional requirements define the overall quality attributes and constraints of the system. They are organized by category following the IEEE Std 830-1998 and FURPS+ classification models. Each requirement includes a measurable metric to allow objective verification.

ID	Category	Requirement	Priority	Metric / Measure
NFR01	Performance	The system shall load any page within 3 seconds under normal usage conditions.	High	Page load time $\leq$ 3 seconds
NFR02	Performance	The system shall return search results for patient name or national ID within 1 second.	High	Search response time $\leq$ 1 second
NFR03	Performance	The system shall complete the appointment booking process and show confirmation within 2 seconds.	High	Booking confirmation time $\leq$ 2 seconds
NFR04	Performance	The secretary dashboard shall load all daily statistics within 2 seconds.	High	Dashboard load time $\leq$ 2 seconds
NFR05	Performance	The system shall handle a minimum of 100 appointment records per day without performance degradation.	Medium	100 daily records with no slowdown
NFR06	Performance	The system shall support at least 50 simultaneous users without performance issues.	Medium	50 concurrent users with no degradation
NFR07	Security	All user passwords shall be stored using a secure hashing algorithm such as bcrypt.	High	0% of passwords stored in plain text
NFR08	Security	The system shall invalidate the user session token after 30 minutes of inactivity.	High	Session timeout = 30 minutes
NFR09	Security	The system shall enforce role-based access control so that each user can only access their own features.	High	100% of pages protected by role verification
NFR10	Security	The system shall ensure that a logged-in patient can only view their own records.	High	0% cross-patient data leakage
NFR11	Security	The system shall validate all form inputs on both client and server side.	High	100% of forms validated on both sides
NFR12	Security	The system shall log all login attempts including successful and failed attempts with timestamp.	High	100% of login attempts logged
NFR13	Security	All sensitive patient medical data shall be encrypted in the database.	High	100% of sensitive fields encrypted
NFR14	Security	All external communications shall use HTTPS protocol.	High	100% of communications over HTTPS
NFR15	Security	The system shall block an account after 5 consecutive failed login attempts.	High	Account locked after 5 failed attempts
NFR16	Usability	The system interface shall be in Persian with right-to-left layout.	High	100% of UI text in Persian RTL format
NFR17	Usability	A new secretary shall complete core tasks within 10 minutes of first use without training.	High	Core tasks completable in $\leq$ 10 minutes
NFR18	Usability	The system shall display clear error messages in Persian for all invalid inputs.	High	100% of errors with descriptive messages
NFR19	Usability	Patient status changes shall be visible immediately without page refresh.	High	Status update visible within 1 second
NFR20	Usability	All critical actions shall require a confirmation dialog before execution.	High	100% of critical actions require confirmation
NFR21	Usability	The system shall be fully usable on mobile screens with minimum width of 375px.	High	Fully functional on screens $\geq$ 375px

ID	Category	Requirement	Priority	Metric / Measure
NFR22	Usability	The patient portal shall guide new users through registration and first booking in under 5 minutes.	Medium	New patient completes first booking in ≤ 5 minutes
NFR23	Usability	Core tasks shall be completable within a maximum of 3 clicks from the main dashboard.	High	Core tasks completable in ≤ 3 clicks
NFR24	Reliability	The system shall maintain 99% availability during clinic working hours.	High	Maximum 1% downtime during working hours
NFR25	Reliability	The system shall recover automatically within 5 minutes after an unexpected crash.	High	Automatic recovery within 5 minutes
NFR26	Reliability	The system shall automatically back up all data every 24 hours.	High	Automatic backup every 24 hours
NFR27	Reliability	The system shall prevent double booking of the same time slot.	High	0% double bookings allowed
NFR28	Reliability	SMS notifications shall be delivered within 5 minutes of the scheduled send time.	Medium	SMS delivery within 5 minutes
NFR29	Portability	The system shall be fully functional on Chrome, Firefox and Safari browsers.	High	100% functionality on Chrome, Firefox, Safari
NFR30	Portability	The system shall be accessible through a web browser without requiring any software installation.	High	Zero installation required for any user
NFR31	Portability	The system shall display correctly on both desktop and mobile devices.	High	Fully responsive on all screen sizes
NFR32	Maintainability	The system code shall be organized into separate modules for each feature.	Medium	Each feature in its own separate module
NFR33	Maintainability	The system shall include technical documentation describing the structure and key functions.	Medium	Documentation covers 100% of main modules
NFR34	Maintainability	The system shall be built with standard HTML, CSS and JavaScript for future extensibility.	Medium	No proprietary frameworks preventing future development
NFR35	Compliance	The system shall comply with patient data privacy regulations.	High	100% compliance with data privacy requirements
NFR36	Compliance	The system shall obtain patient consent before storing their medical data.	High	100% of patients shown consent form at registration
NFR37	Capacity	The system shall be capable of storing records for a minimum of 1000 patients.	High	Minimum 1000 patient records supported
NFR38	Capacity	The system shall retain appointment and treatment history for a minimum of 5 years.	Medium	5 years of historical data retained
NFR39	Capacity	The system shall support at least 3 concurrent staff users simultaneously.	High	3 concurrent staff users with no degradation

6. User Stories

User stories capture requirements from the perspective of the end user. They follow the standard Agile format: As a [user], I want to [action] so that [benefit]. Stories are organized by sprint and linked to their corresponding Jira Epic.

ID	Epic	Summary	User Story	Priority	Sprint
US01	Secretary	View today's appointment list on dashboard	As a secretary, I want to see the list of today's patients on the dashboard so that I can quickly manage the daily work schedule.	High	Sprint 1
US02	Secretary	Search for a patient by name or national ID	As a secretary, I want to search for a patient by name or national ID so that I can quickly access their file.	High	Sprint 1
US03	Secretary	Register a new patient file	As a secretary, I want to register a new patient file so that their information is saved for future visits.	High	Sprint 1
US04	Secretary	Change patient status during visit	As a secretary, I want to change the patient status between Waiting, In Doctor's Room and Visit Completed so that I always know the current situation.	High	Sprint 1
US05	Secretary	Record patient payment amount	As a secretary, I want to record the amount paid by the patient so that the daily accounts are accurate.	High	Sprint 1
US06	Secretary	View patient medical alerts before appointment	As a secretary, I want to see important medical alerts for each patient so that I can warn the doctor before the visit begins.	High	Sprint 1
US07	Secretary	View recent activity log	As a secretary, I want to see a log of recent activities on the dashboard so that I can track what has happened during the day.	Medium	Sprint 1
US08	Secretary	Follow secretary daily checklist	As a secretary, I want to see a daily checklist of important tasks so that I don't forget critical steps.	Medium	Sprint 1
US09	Secretary	Contact patient from appointment list	As a secretary, I want to contact a patient directly from the appointment list so that I can remind them or follow up.	Low	Sprint 1
US10	Secretary	View daily statistics summary cards	As a secretary, I want to see summary cards showing total appointments, waiting patients, completed visits and daily income.	High	Sprint 1
US11	Authentication	Login with email and password	As a user, I want to log in with my email and password so that I can access the system securely based on my role.	High	Sprint 2
US12	Authentication	Logout from the system securely	As a user, I want to log out from the system so that my account is protected when I finish my session.	High	Sprint 2
US13	Authentication	Reset forgotten password via email	As a user, I want to reset my password via email so that I can regain access if I forget my password.	High	Sprint 2
US14	Authentication	See different dashboard based on role	As a user, I want to be redirected to my own dashboard after login based on my role so that I only see relevant features.	High	Sprint 2
US15	Authentication	Admin creates new staff accounts	As a manager, I want to create new accounts for staff members so that they can access the system with appropriate permissions.	High	Sprint 2
US16	Authentication	Admin deactivates staff accounts	As a manager, I want to deactivate a staff member's account so that former employees can no longer access the system.	High	Sprint 2

ID	Epic	Summary	User Story	Priority	Sprint
US17	Authentication	Account locks after failed login attempts	As a system, I want to lock an account after 5 consecutive failed login attempts so that unauthorized access is blocked.	High	Sprint 2
US18	Authentication	Session expires after inactivity	As a system, I want to automatically log out a user after 30 minutes of inactivity so that patient data is protected.	High	Sprint 2
US19	Patient Records	Register patient special medical conditions	As a secretary, I want to register a patient's special medical conditions when creating their file so that the doctor is informed before treatment.	High	Sprint 3
US20	Patient Records	View full patient file	As a secretary, I want to open a patient's full file so that I can view all their information including contact details and medical history.	High	Sprint 3
US21	Patient Records	View patient treatment history	As a dentist, I want to view the patient's treatment history so that I can make a more accurate treatment decision.	High	Sprint 3
US22	Patient Records	Record type of treatment performed	As a dentist, I want to record the type of treatment performed so that the patient's medical history stays complete.	High	Sprint 3
US23	Patient Records	View medical alerts and allergies	As a dentist, I want to see alerts about a patient's drug allergies so that I can perform safer treatment.	High	Sprint 3
US24	Patient Portal	Register as a new patient online	As a patient, I want to register an account online so that I can book appointments without visiting the clinic.	High	Sprint 4
US25	Patient Portal	Login to patient portal	As a patient, I want to log in to my account so that I can access my appointments and medical records securely.	High	Sprint 4
US26	Patient Portal	Book an appointment online	As a patient, I want to book an appointment online by selecting a date and available time slot.	High	Sprint 4
US27	Patient Portal	View upcoming appointments	As a patient, I want to see a list of my upcoming appointments so that I can keep track of my scheduled visits.	High	Sprint 4
US28	Patient Portal	Cancel appointment online	As a patient, I want to cancel an upcoming appointment online so that the time slot is freed for other patients.	High	Sprint 4
US29	Patient Portal	View treatment history	As a patient, I want to view my past treatment history so that I can track all dental procedures I have had.	Medium	Sprint 4
US30	Patient Portal	View payment history and outstanding balance	As a patient, I want to view my payment history and any outstanding balance.	Medium	Sprint 4
US31	Patient Portal	Receive booking confirmation	As a patient, I want to receive a confirmation message after booking so that I have proof of my appointment.	High	Sprint 4
US32	Notifications	Send SMS reminder before appointment	As a system, I want to send an SMS reminder to the patient 24 hours before their appointment.	Medium	Sprint 5
US33	Notifications	Send email confirmation after booking	As a system, I want to send an email confirmation immediately after booking so the patient has a written record.	Medium	Sprint 5

ID	Epic	Summary	User Story	Priority	Sprint
US34	Notifications	Send payment reminder	As a system, I want to send a payment reminder to patients with an outstanding balance.	Medium	Sprint 5
US35	Notifications	Notify secretary of new online booking	As a secretary, I want to receive a notification when a patient books online so that I can update the daily schedule.	High	Sprint 5
US36	Doctor Panel	View patient allergy warnings	As a dentist, I want to see clear warnings about drug allergies before starting treatment so I can choose safe options.	High	Sprint 6
US37	Doctor Panel	Add treatment plan for patient	As a dentist, I want to create a treatment plan so that proposed services are recorded and visible to the secretary.	High	Sprint 6
US38	Doctor Panel	View today's patient schedule	As a dentist, I want to see my schedule for today so that I can prepare for each patient's treatment.	High	Sprint 6
US39	Doctor Panel	Issue prescription for patient	As a dentist, I want to issue a prescription after a visit so that the patient can purchase required medication.	High	Sprint 6
US40	Doctor Panel	View all prescriptions issued to patient	As a dentist, I want to view all prescriptions issued to a patient so I avoid prescribing conflicting medications.	High	Sprint 6
US41	Manager Reports	View daily income report	As a manager, I want to view the daily income so that I can review the clinic's financial performance.	High	Sprint 7
US42	Manager Reports	View daily patient and service summary	As a manager, I want to see the number of patients and services recorded to evaluate daily performance.	High	Sprint 7
US43	Manager Reports	View cash vs card payment breakdown	As a manager, I want to see a breakdown of payments by type so that I can reconcile daily accounts.	Medium	Sprint 7
US44	Manager Reports	Generate monthly financial report	As a manager, I want to generate a monthly financial report to review total income and outstanding balances.	Medium	Sprint 7
US45	Appointment	View patient journey flow	As a secretary, I want to see the full patient journey from arrival to payment so I can manage the workflow efficiently.	High	Sprint 8
US46	Appointment	Track pending payments	As a secretary, I want to see which patients have pending payments so that I can follow up before they leave.	High	Sprint 8
US47	Appointment	Add insurance information to patient file	As a secretary, I want to add a patient's insurance information so that the correct billing can be applied.	High	Sprint 8
US48	Appointment	Calculate payment after insurance deduction	As a secretary, I want the system to calculate the patient's payment after insurance coverage automatically.	High	Sprint 8

7. General Usage Scenario

The following scenario demonstrates the typical daily workflow of the system from the perspective of all key users:

Step 1 — Secretary arrives and opens the dashboard

The secretary logs in and is redirected to the secretary dashboard. They see today's appointment list, daily statistics cards showing total appointments, patients waiting and current income, and a checklist of daily tasks.

Step 2 — New patient arrives

A new patient arrives. The secretary searches for them by name or national ID. If they are not found, a new patient file is created with their name, national ID, phone number, date of birth and any special medical conditions or allergies.

Step 3 — Patient status is updated

The patient is added to today's list with status Waiting. When they are called into the treatment room, the secretary changes their status to In Doctor's Room. The status badge updates immediately without page refresh.

Step 4 — Dentist reviews the patient

The dentist opens the patient file and reviews their treatment history and any medical alerts. After the examination, the dentist records the service performed and issues a prescription if needed. A treatment plan is created for the patient's next visit.

Step 5 — Payment is recorded

After the visit is complete, the secretary records the payment amount. The system calculates any insurance deductions automatically. A receipt is generated for the patient. The patient's status is changed to Visit Completed.

Step 6 — Patient books their next appointment online

The patient logs into the patient portal and books their next appointment by selecting a date and available time slot. They receive an email confirmation immediately. An SMS reminder will be sent 24 hours before the appointment.

Step 7 — Manager reviews end of day reports

At the end of the day, the manager logs in and views the daily report showing total income, number of patients seen, services performed, and a breakdown of cash vs card payments. Outstanding patient balances are also visible for follow-up.

8. Prototype Description

8.1 Prototype Overview

The HTML prototype for this project demonstrates the user experience of the system across all eight epics defined in the Jira backlog. The prototype does not contain a backend or real data logic — it is a visual and interactive representation of the system screens and user flow, built using static HTML, CSS and JavaScript to demonstrate requirements coverage and UX design.

Sprint 1 (Secretary Dashboard) was the first epic to reach a fully interactive demo. The prototype has since been extended with a dedicated screen for every remaining epic — authentication and role-based dashboards, patient records, the doctor panel, the patient portal, the notification center, finance and manager reporting — together with a requirements-coverage page that maps each screen back to its corresponding user stories and functional requirements for the course presentation.

8.2 Prototype Screens

Secretary Dashboard (index.html)

- Hero section with patient journey flow showing 4 stages: Search/Register, Waiting, In Room, Payment
- Daily statistics cards: Total appointments, Patients waiting, Completed visits, Daily income
- Live kanban-style board of the patient flow (Waiting, In Room, Awaiting Payment, Completed)
- Side panel with recent activity log and daily checklist

Today's Appointments (appointments.html)

- Full appointment table with a search box filtering by name, national ID or service
- Action buttons: Change Status, View File, Record Payment, Contact Patient
- Highlighted medical-alert notes per patient row
- Reference list of the User Stories demonstrated on the page

Patient Registration Form (modal, available from the dashboard)

- Modal form for registering new patients
- Fields: Full name, National ID, Phone number, Service type, Medical condition, Appointment time, Notes
- Submit creates an entry in today's appointment list

Login & Role-Based Dashboards (login.html)

- Demo login form with selectable role tabs: Secretary, Dentist, Manager, Patient
- A role-specific preview panel describing the dashboard each role lands on after login
- Security-control checklist: account lockout after failed attempts, session timeout, password recovery, login logging

Patient Records (patients.html)

- Patient profile card with avatar, name, national ID, contact details and active-file tags
- Prominent medical-alert banner for drug allergies and chronic conditions
- Tabbed view for overview, treatment history, payments, notes and prescriptions
- Treatment history table with date, service, dentist and cost

Doctor Panel — Treatment & Prescriptions (doctor.html)

- Today's doctor schedule list with linked patient name and service
- Sample prescription panel with medication, dosage and an allergy warning

- Treatment-plan table showing proposed service, status and priority per patient
- Placeholder card for a future radiography / OPG image request

Patient Portal & Online Booking (portal.html)

- Four-step booking timeline: online registration, free-slot selection, booking confirmation, appointment management
- Sample booking form with doctor and available time-slot selection
- Upcoming-appointments list with a cancel action
- Payment history and outstanding-balance summary visible to the patient

Notification Center (portal.html#notifications)

- Sample SMS appointment-reminder card sent 24 hours in advance
- Sample email booking-confirmation card
- Internal “new online booking” alert shown on the secretary's dashboard

Finance Section (finance.html)

- Payment breakdown showing cash, card and insurance amounts
- Sample insurance-coverage calculation (service cost, coverage %, patient share)
- Outstanding-balance tracking and a receipt-generation button

Manager Reports (reports.html)

- Daily patient count and completed-service summary
- Income report with a cash vs card payment-type breakdown
- Preview action for a future monthly financial report

Requirements Coverage Map (requirements.html)

- Six coverage cards linking each prototype area to its User Story and Functional Requirement ranges
- Used during the course presentation to show traceability from prototype screens back to this document

8.3 Prototype to Requirements Mapping

Prototype Element	User Stories Covered	Functional Requirements
Dashboard appointment table	US-01, US-10	FR-01, FR-08
Search box	US-02	FR-02, FR-03
Register New Patient form	US-03, US-19	FR-04, FR-27
Change Status button	US-04	FR-05, FR-06
Record Payment button	US-05	FR-07
Medical alert display	US-06, US-23	FR-12, FR-29
Activity log panel	US-07	FR-09
Daily checklist	US-08	FR-10
Statistics cards	US-10	FR-08
Treatment history table	US-21, US-22	FR-33, FR-34
Finance section	US-41, US-43	FR-62, FR-64
Daily reports section	US-42	FR-63
Login form & role tabs	US-11, US-14	FR-17, FR-18

Prototype Element	User Stories Covered	Functional Requirements
Security-control checklist	US-17, US-18	FR-21, FR-22
Doctor schedule panel	US-38	FR-36
Prescription panel	US-39, US-40	FR-38, FR-39
Treatment-plan table (doctor)	US-37	FR-37
Booking timeline & reservation form	US-26, US-31	FR-44, FR-51
Upcoming appointments / cancel action	US-27, US-28	FR-46, FR-47
Notification cards (SMS / email / internal)	US-32, US-33, US-35	FR-54, FR-55, FR-57
Patient profile & medical info tabs	US-20, US-23	FR-14, FR-28, FR-29
Requirements coverage map	Presentation aid — all epics	Traceability only, no single FR

9. Agile Sprint Plan

The project is organized into 8 sprints following the Scrum methodology. Each sprint corresponds to one Epic in Jira and represents a logical grouping of related features. Sprint 1 was the first to be prototyped and demonstrated; the prototype has since been extended to cover every remaining sprint.

Sprint	Name	Goal	Status	Epic
1	Secretary Dashboard	Complete secretary dashboard prototype with patient status management and daily income tracking	Done	Secretary Dashboard Management
2	Login & Authentication	Implement secure login, role-based access control and password management for all user types	Prototype Demo Ready	Login and Authentication
3	Patient Records	Build complete patient file management including registration, search and medical history	Prototype Demo Ready	Patient Records Management
4	Patient Portal	Allow patients to register, login, book appointments and view their own records online	Prototype Demo Ready	Patient Portal
5	Notifications	Implement SMS and email notifications for appointment reminders and payment alerts	Prototype Demo Ready	Notifications and Reminders
6	Doctor Panel	Build doctor dashboard with treatment recording, prescription management and patient history	Prototype Demo Ready	Doctor Panel
7	Manager Reports	Implement financial reports, insurance billing and monthly performance analytics	Prototype Demo Ready	Manager Reports and Admin
8	Appointment Management	Complete online appointment booking system with time slot management and conflict prevention	Prototype Demo Ready	Appointment Management

Note: “Prototype Demo Ready” indicates that a representative, interactive UI mock-up for the epic has been built and demonstrated in the HTML prototype (see Section 8). It does not indicate that backend logic, database persistence or real authentication has been implemented — full functional development for Sprints 2–8 remains planned for a future software engineering course, consistent with Section 1.4.

10. Conclusion

This document presents a complete system analysis and design for a Dental Clinic Management System. The project includes a full requirements specification with 72 functional requirements and 39 non-functional requirements, 48 user stories organized across 8 sprints, a detailed Jira project board, and an interactive HTML prototype demonstrating representative screens and core user flows for every epic — from the secretary dashboard and patient records to the doctor panel, patient portal, notifications, finance and manager reporting.

The system is designed with future development in mind. Sprint 1 has been implemented as a fully interactive prototype, and Sprints 2–8 now also have representative screens demonstrated in the HTML prototype; full backend implementation across all sprints remains the roadmap for a future software engineering course. The modular design using standard HTML, CSS and JavaScript ensures that the system can be extended and converted into a fully functional web application.

The project demonstrates a complete system analysis workflow covering problem identification, stakeholder analysis, requirements engineering, user story writing, Agile sprint planning, and UI prototyping — all key competencies of a systems analysis and design course.

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